

When the Judge Acts: Auditing VLM-Guided Image Selection on Culturally Situated Prompts

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Code and project page —

<https://github.com/seochan99/JudgeActs>

Abstract

Vision-language models (VLMs) increasingly act as selectors: given several generated images, they decide which one a user receives. Such judges are usually validated by how well their scores agree with human ratings, which says little about the images they actually return. We audit VLM selection as an action. On 300 culturally situated prompts, we compare each returned image with released human ratings the judge never sees and with exact random choice on the same pool, under three rotations of candidate order. We make three contributions. (1) A matched-baseline audit protocol for selectors, released with a deterministic artifact. (2) Evidence that presentation strongly shapes small judges: a 4B-parameter judge picks the first-listed image in 48.8% of calls (27.8% expected), changes its choice under reordering on 59.7% of prompts, and beats random only slightly (gain 0.039, 95% CI [0.016, 0.062]), less than a CLIP similarity scorer (0.069). (3) An audit of agreement gates on what they keep and what they reject: requiring agreement across orders keeps 40.3% of prompts, on which the judge beats random by 0.141, whereas agreement with a weaker second judge keeps below-random choices. Both selectors reduce missing cultural expectations but slightly raise stereotype ratings, so cultural errors must be audited separately from alignment.

1 Introduction

A vision-language model (VLM) that selects one of several generated images does more than report a quality score. Its choice determines which candidate a user would receive. The practical question is whether that choice improves the available outputs, not just whether the judge’s scores correlate with human ratings.

For culturally situated requests, visual polish and recognizable symbols can be poor guides to prompt satisfaction. An image may include a requested object while misrepresenting the social context around it. A selector can favor that image even when another candidate better satisfies the request. Evaluating this choice requires ratings that did not come from the selector itself.

CulturalFrames provides generated images and human ratings that distinguish missing explicit expectations, missing implicit expectations, and stereotypes. It also reports weak

agreement between automatic metrics and human judgments (Nayak et al. 2025). Related work examines cultural representation in generation and editing (Seo et al. 2025). We use the public CulturalFrames release to study a different decision: choosing which existing image to return.

For each prompt, an open-weight VLM sees three or four candidate images and selects one. It receives no generator names, country metadata, or human ratings. We retrieve the selected image’s released ratings and compare its score with the exact expectation of random selection and the highest available annotation score. The candidate pool is fixed, so differences between policies arise from selection rather than generation.

The primary outcome is *selection regret*, measured from the mean human prompt-alignment rating. It is the gap between the highest-rated available candidate and the selected image. Prompt alignment measures request satisfaction, not cultural accuracy alone. We therefore report stereotype and missing-expectation ratings separately. We also measure how often the selected score falls below random expected utility. We call this the *below-mean rate* (BMR). It is a score shortfall relative to random choice, not a measure of harm to users.

We ask four questions. (RQ1) Does VLM selection improve prompt alignment relative to random choice? (RQ2) How do regret and below-baseline selection vary across country groups? (RQ3) How does selection behave when explicit or implicit error ratings differ across candidates? (RQ4) Does agreement across candidate orders identify selections with lower regret?

The audit connects each judge’s choice to an independently collected outcome rating. We contribute a matched-baseline audit protocol and three findings:

- **A matched-baseline selection audit.** Each choice is compared with the exact expectation of random selection on the same candidate pool, so gains, below-mean selections, and cultural-error ratings are paired rather than pooled.
- **Presentation strongly shapes small judges’ choices.** Both judges strongly prefer early presentation slots; the same prompt can yield the worst or the best available image depending only on candidate order (Figure 1).
- **Agreement rules must be audited on what they reject.** Order unanimity separates prompts where the judge beats

random from those where it falls below random, whereas agreement with a weaker second judge retains below-random choices.

2 Related Work

Cultural representation and evaluation. CulturalFrames supplies culturally situated prompts, multiple generated images, and ratings of missing expectations and stereotypes (Nayak et al. 2025). Its analysis compares automatic metrics with human ratings of individual images. Our analysis instead evaluates choices within a candidate set. Seo et al. (2025) study cultural representation in image generation and editing; we study selection among existing outputs.

Chang and Chen (2026) train a cultural reward model on preferences derived from CulturalFrames and evaluate pairwise agreement. This is closely related evidence that cultural image preferences can be learned. We do not propose a competing reward model; we audit frozen, prompted selectors and the requests retained by their abstention rules.

VLM judges. Zou et al. (2026) show that multimodal judges can favor informative answers without adequately using visual evidence. Park et al. (2026) examine conflicts between textual narratives and perception, and train judges to ground their decisions more reliably. Both concern tasks distinct from our candidate-image selection setting. They motivate checking selected outcomes against independent

ratings, but do not establish the mechanism behind the errors measured here.

K-Sort Eval uses multi-image VLM rankings, posterior correction, and adaptive comparisons to evaluate visual generators (Li et al. 2026). Our study asks about the selected image within each prompt’s candidate pool, rather than the accuracy or efficiency of a generator leaderboard. We do not claim that comparative image judging or best-of-pool selection is new.

Position bias in model judges. Language-model judges are sensitive to the order in which options are presented: pairwise judges favor a position (Zheng et al. 2023; Wang et al. 2024; Shi et al. 2025), and multiple-choice accuracy shifts with option order (Pezeshkpour and Hruschka 2024). Multimodal judges also show position and consistency effects on vision-language benchmarks (Chen et al. 2024), over-score earlier sentences and prefer their own captions when assessing image-text alignment (Saito et al. 2025), and can be swayed by visual manipulations unrelated to quality (Hwang et al. 2025). Proposed mitigations include swapping or averaging over orders (Zheng et al. 2023; Wang et al. 2024) and de-biased judging protocols (Asaria, Salomone, and Gandhi 2026). VLM evaluators are also used to audit generated images at scale, with human validation on a subset (Seo et al. 2026). We measure the same phenomenon when the judge’s output is an action, and ask how it interacts with agreement-based abstention.

Judgments used as decision rules. Yao et al. (2026) investigate multimodal reinforcement learning with imperfect rewards, where gains on a proxy can diverge from task performance. WORLDVIEW examines cultural stereotyping introduced during prompt revision (Urman et al. 2026). We isolate a later decision, selecting an existing candidate. The experiment does not optimize generators against a reward model or simulate a sequential feedback process.

Selective prediction. Risk-coverage analysis evaluates prediction with a reject option (El-Yaniv and Wiener 2010; Geifman and El-Yaniv 2017). We apply it to image selection, using agreement across orders or judges as acceptance rules. These rules yield measured operating points, not calibrated confidence scores or risk guarantees.

3 Problem and Measures

Selection task. Let p denote a prompt and $C_p = \{x_{p1}, \dots, x_{pm_p}\}$ its pool of $m_p \in \{3, 4\}$ candidate images, already generated and held fixed. A *selection policy* π receives the prompt and the images, shown in a given order with labels, and either returns the index $\pi(p) \in \{1, \dots, m_p\}$ of one candidate or abstains. We always record the chosen image’s identity, not its label, so the same image can carry the label “A” in one order and “C” in another.

Human reference utility. Each candidate x_{pi} has n_{pi} released prompt-alignment ratings $a_{pir} \in \{0, 0.5, 1\}$, one per annotator r . Its utility is the mean rating,

$$h_{pi} = \frac{1}{n_{pi}} \sum_{r=1}^{n_{pi}} a_{pir} \in [0, 1]. \quad (1)$$

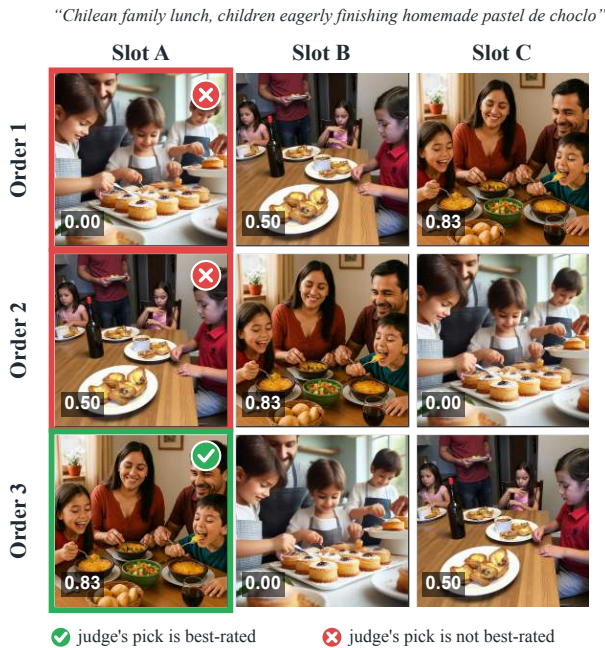


Figure 1: Same request, same images, three orders. Each row shows the three candidates in one presentation order, with mean human prompt-alignment ratings. Qwen3-VL-4B picks the image in slot A every time (framed), so its returned image is rated 0.00, 0.50, or 0.83 depending only on order. Images: CulturalFrames (Nayak et al. 2025).

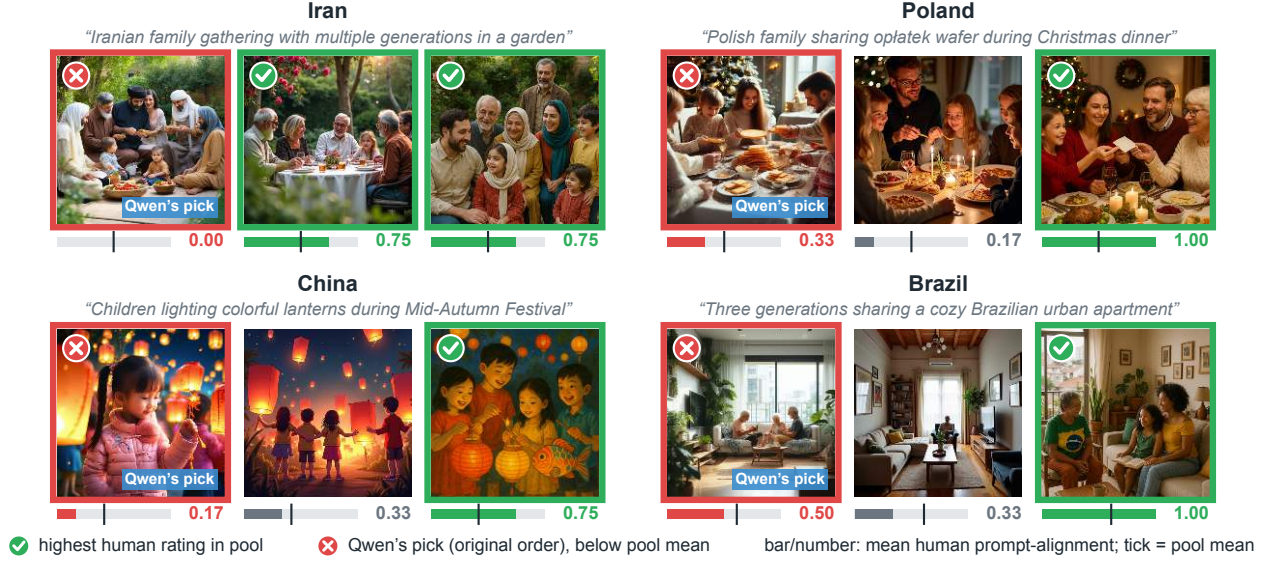


Figure 2: VLM judges can return below-average images when a clearly better candidate exists. Each group is one CulturalFrames prompt (Nayak et al. 2025); numbers are mean human prompt-alignment ratings. Red: Qwen3-VL-4B’s original-order choice, below the pool mean. Green: highest-rated candidate. Prompts follow a fixed rule applied without viewing images (Appendix D); they illustrate the failure mode, not its frequency (Figure 4).

Judges never see these ratings; they are joined to the chosen image only after selection.

Baselines. Choosing uniformly at random has expected utility $\bar{h}_p = \frac{1}{m_p} \sum_{i=1}^{m_p} h_{pi}$, which we compute exactly rather than by simulation. The *annotation oracle* picks the highest-rated candidate, with utility $h_p^* = \max_i h_{pi}$; it is a reference point, not a deployable selector.

Outcome measures. For a policy that acts on prompt p and returns image $x_{p,\pi(p)}$, with utility $h_{p,\pi(p)}$, we report three quantities:

$$\text{regret: } R_p(\pi) = h_p^* - h_{p,\pi(p)}, \quad (2)$$

$$\text{gain over random: } G_p(\pi) = h_{p,\pi(p)} - \bar{h}_p, \quad (3)$$

$$\text{below-mean event: } B_p(\pi) = \mathbb{1}\{h_{p,\pi(p)} < \bar{h}_p\}. \quad (4)$$

Regret is the primary outcome: zero means the judge returned a best-rated image. Gain is positive when the judge beats random choice on the same pool, and the two are linked by $G_p = (h_p^* - \bar{h}_p) - R_p$. Averaging B_p over prompts gives the *below-mean rate* (BMR), a score shortfall relative to random, not a measure of user harm. Random choice has its own below-mean probability, the share of candidates rated below $\bar{h}_p$, which we use as its matched comparison. We also report how often the policy returns a best-rated image, counting ties. Every average is taken over the set A_π of prompts on which the policy acts, and each policy is compared with random choice on exactly those prompts.

Cultural-error outcomes. CulturalFrames also releases, per image, the share of annotators who flag a stereotype, a missing explicit expectation, or a missing implicit expectation. For each of these error rates we report the chosen image’s value minus the random-choice average over the same

pool, so that negative values mean fewer errors than chance. These outcomes are kept separate from alignment and are not combined into a composite score. A pool is *sensitive* to an error type when that error rate differs across its candidates, so that selection can change it; we report results on each sensitive subset (Table 2). As a secondary utility we also use overall satisfaction, rescaled from its 1–5 scale to $[0, 1]$.

Presentation orders and agreement gates. Each pool is shown in $K = 3$ cyclic orders $k \in \{1, 2, 3\}$ (order 1 is canonical, orders 2 and 3 rotate it). Let $\pi_k(p)$ be the index chosen under order k and

$$q_p = \frac{1}{K} \max_i |\{k : \pi_k(p) = i\}| \quad (5)$$

the share of orders that agree on the most frequent choice. A *gate* with threshold τ acts only when $q_p \geq \tau$ and then returns that majority choice: $\tau = 1$ is *unanimity*, $\tau = 2/3$ requires two of three, and $\tau = 1/3$ always acts. The *cross-model gate* acts only when two judges make the same order-1 choice. For every gate we report coverage $|A_\pi|/N$ over the $N = 300$ prompts, the outcomes on the prompts it keeps, and, because a gate can look good simply by keeping easy pools, the same outcomes for random choice and for the ungated judge on both the kept and the rejected prompts.

Country groups. Each prompt belongs to one of ten country contexts; we report group means of regret and gain with their random baselines, and *worst-group regret*, the largest group mean. Country labels identify dataset contexts, not homogeneous cultural preferences.

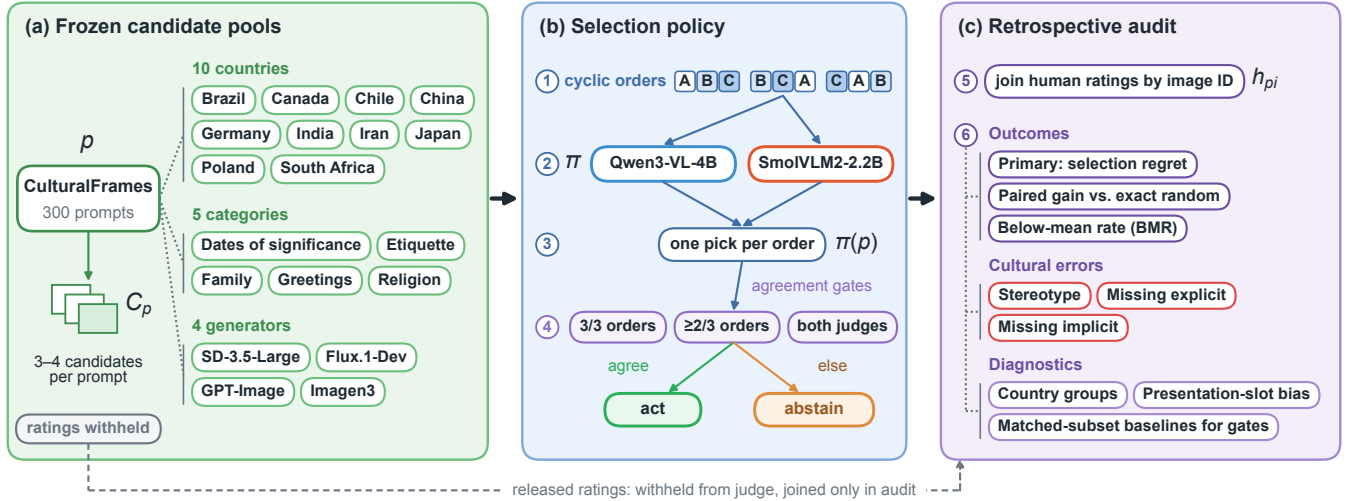


Figure 3: Audit protocol. (a) Frozen candidate pools from CulturalFrames: culturally situated prompts with three or four images from four generators. (b) Selection: each judge sees only the prompt and the images, in three cyclic orders; agreement gates decide whether to act or abstain. (c) Retrospective audit: released human ratings, never shown to the judge, are joined by image identifier after selection and compared with exact random selection on the same pools.

4 Experimental Setup

4.1 Dataset and Frozen Splits

We use the independently released CulturalFrames dataset (Nayak et al. 2025). Image rows are joined to annotation records by exact image identifier, with prompt, country, and generator-suffix checks. We decode every image, check repeated identifiers and encoded-image content, and require at least three valid, annotated candidates. For repeated prompt text within a country we retain the lexicographically first base identifier before splitting, preventing duplicated requests from crossing splits. Missing annotations are never imputed. The provenance manifest pins the public dataset and model revisions.

The release contains 3,637 image rows and 3,577 image annotation records, comprising 10,412 individual ratings. 60 image rows have no annotation record (1.6%). Filtering leaves 963 eligible sets. Removing 2 repeated requests within countries leaves 961 unique prompts: 313 three-candidate and 648 four-candidate sets. The main split has 100 three-candidate and 200 four-candidate sets; 11 sets have identical candidate alignment scores.

We create a development split of three prompts per country and a disjoint main split of 30 per country, using seed 20261002. Split membership depends only on a deterministic hash of prompt identifiers, without judge outputs or annotation values. The independent analysis unit is a prompt candidate set, not an image or one permutation call. The equal country allocation defines a balanced-country estimand; it does not reproduce a real deployment distribution. Remaining eligible sets form a reserve, and the main split is not expanded in response to its results.

4.2 Open-Weight Judges

The primary judge is Qwen3-VL-4B-Instruct (Qwen Team 2025); SmolVLM2-2.2B-Instruct is the secondary judge (Marafioti et al. 2025; HuggingFaceTB 2025). Model cards, revisions, dependency versions, and inference settings are stored with the artifact. We run locally with greedy decoding, a 64-token output limit, and no requested explanation. Qwen images fit an aspect-preserving 448-pixel bounding box. Smol uses its native 384-pixel global view with image tiling disabled to bound memory consumption. These configurations provide different visual token budgets and limit interpretation of cross-model differences as differences in model capability alone.

Every request interleaves a candidate label with its image. The instruction asks for explicit prompt satisfaction, implicit contextual consistency, avoidance of unsupported stereotypes, and visual integrity. It warns against rewarding cultural symbols merely for their number or recognizability. Only the original prompt supplies cultural context. Source model names, category labels, country metadata, and human annotations do not enter the request.

For Qwen the output schema is a JSON object with `choice` and `ranking`. The ranking must contain every supplied label exactly once, and its first label must equal the choice. Smol receives the same evaluation criteria but a choice-only JSON contract, selected during development because it could not reliably satisfy the full-ranking format. Its adapter can normalize a single explicit label in a one-field structured object; raw JSON and exact-schema validity are tracked separately from recovered decision validity. Rankings are not used by any outcome metric. These interface differences limit comparisons of formatting reliability. We record raw text, parsed output, label-to-image mapping, input tensor shapes, model revision, elapsed time, and request

hashes. Unit tests verify regret extrema, the analytic random expectation, tie behavior, annotation aggregation, and identity preservation under permutation.

4.3 Candidate Order and Invalid Outputs

Each prompt receives three distinct cyclic presentations: its canonical identifier order and two deterministic rotations. Canonical order sorts candidate identifiers, which usually places the same generator first; the rotations break this link, so generator effects are assessed over all three orders. The instruction, images, and decoding settings remain fixed. Thus disagreement measures sensitivity to these presentation changes rather than sampled decoding variation or prompt paraphrases. Three rotations do not exhaust all orders, especially for four candidates.

The original-order decision is the primary direct policy. No main-run prompt edits are permitted. An unresolved parse failure produces an abstention and remains in the output log. Qwen uses strict schema validation; Smol uses its predeclared single-label normalization, with every recovery flagged. A runtime exception permits one retry with identical inputs; invalid text is not repeatedly sampled until valid. Utility and regret are conditional on valid decisions, with coverage relative to the full frozen split reported alongside them. We compare gain to random on the same acted-on prompts. We never assign failed requests a favorable candidate or silently remove their denominator.

4.4 Selection with Abstention

The gate thresholds $\tau \in \{1/3, 2/3, 1\}$ and the cross-model rule were declared before main inference. A gate acts only when all three order calls parse; ties in the majority are broken by canonical identifier order, so the action never depends on human ratings. Abstention leaves a request unresolved: we do not assume a human fallback or a zero-risk outcome, and we report coverage next to every selective estimate. Because Smol is a weaker judge (Section 5), cross-model results are conditional on that pairing.

4.5 Statistical Analysis

The unit of analysis is a prompt with its candidate pool; the three order calls for a prompt are never resampled independently. We report 95% intervals from 10,000 prompt-level bootstrap resamples (seed 20261002) for means and paired differences, and Wilson intervals for a single policy’s BMR. Intervals condition on the released annotation means. Secondary cultural-error intervals are pointwise, and we additionally report Bonferroni-adjusted intervals across the three error types. Analyses of presentation position, of the prompts that gates reject, and all ablations in Section 5.5 were added after main inference and are labeled exploratory. Appendix M gives further estimation details.

5 Results

5.1 Direct Selection

Table 1 and Figure 4a summarize the direct policies. Qwen returns a valid original-order choice for all 300 prompts. Its mean alignment is 0.770, against 0.730 for random choice

and 0.929 for the oracle, a gain of 0.039 (95% CI [0.016, 0.062]). Mean regret is 0.160 [0.136, 0.184]; Qwen returns a best-rated image for 51.0% of prompts and a below-mean image for 36.3%, against 43.1% for random choice (paired difference -0.067, 95% CI [-0.118, -0.016]).

Smol does worse than random: gain -0.073 [-0.093, -0.052], regret 0.270, and BMR 60.3%. It rarely follows the output format (Appendix Table 10), and its interface and image resolution differ from Qwen’s, so we treat it as a weak secondary judge rather than as a capability comparison. Appendix Table 6 shows that the conclusions hold with a below-mean margin of 0.05, with the median in place of the mean, and with overall satisfaction as the utility.

5.2 Presentation Position

Both judges favor early slots (Figure 4c). Qwen picks slot A in 48.8% of its 900 calls, where uniform choice given the pool sizes would give 27.8% ($\chi^2_3 = 219, p < 10^{-10}$); Smol places 93.3% of its choices in slots A or B. Consistent with this, Qwen’s choice changes across the three orders for 59.7% of prompts. Figure 1 shows the extreme case: the judge returns whatever image is shown first.

The judge still responds to content. Canonical order places one generator first in most pools, so we pool all three rotations, which spreads each generator across slots: Qwen then chooses images from the generator with the highest mean human rating in 57.9% of calls, against 27.2% under uniform choice (Appendix Table 4). Its choices mix a content signal with a strong slot preference.

5.3 Agreement Gates: What They Keep and Reject

A judge that follows position alone picks a different image in each rotation, so it can never be unanimous. Unanimity keeps 121 prompts (40.3%). On them Qwen’s gain over random is 0.141 [0.110, 0.173] and regret 0.074; on the 179 rejected prompts its gain is -0.030 [-0.057, -0.002], below random (Figure 4b). The kept prompts are not easier: their candidates differ more in rating than the rejected ones (mean best-minus-worst range 0.454 versus 0.404), leaving more room for error. The gate only filters prompts; on kept prompts the choice is unchanged.

Voting over the three orders instead of abstaining, a simple form of order averaging, raises the gain on all prompts from 0.039 to 0.065 [0.043, 0.086].

Agreement between judges behaves differently. Qwen and Smol agree on 59 prompts (19.7%), where Qwen’s gain is -0.060 [-0.111, -0.011]; on the 241 prompts where they disagree it is 0.064 [0.039, 0.087]. Two judges that both prefer early slots can agree for reasons unrelated to content, so this gate keeps the wrong prompts. The result is specific to pairing with a weak judge. Appendix Table 5 lists every gate with its complement.

5.4 Comparison with a Content-Only Selector

Would a simpler scorer that cannot be influenced by order do as well? We score each candidate by CLIP prompt-image

Policy	<i>n</i>	Cov. (%)	Align.	Regret [95% CI]	Gain over random [95% CI]	BMR (%)	Best (%)	Worst
Random	300	100.0	0.730	0.199 [0.186, 0.213]	-0.000 [-0.000, 0.000]	43.1	41.9	0.266
Qwen	300	100.0	0.770	0.160 [0.136, 0.184]	0.039 [0.016, 0.062]	36.3	51.0	0.275
Qwen, majority of 3 orders	300	100.0	0.795	0.135 [0.114, 0.156]	0.065 [0.043, 0.086]	31.7	54.3	0.186
Qwen, unanimous gate	121	40.3	0.854	0.074 [0.050, 0.100]	0.141 [0.110, 0.173]	16.5	71.1	0.115
Smol	297	99.0	0.659	0.270 [0.243, 0.298]	-0.073 [-0.093, -0.052]	60.3	29.0	0.400
Cross-model gate	59	19.7	0.690	0.249 [0.187, 0.316]	-0.060 [-0.111, -0.011]	54.2	33.9	0.490
CLIP similarity	300	100.0	0.799	0.130 [0.109, 0.153]	0.069 [0.047, 0.090]	28.0	55.7	–
Oracle	300	100.0	0.929	0.000 [0.000, 0.000]	0.199 [0.186, 0.213]	0.0	100.0	0.000

Table 1: Selection outcomes on the 300 main prompts (95% prompt-level bootstrap intervals). Cov.: share of prompts on which the policy acts. Align.: mean prompt-alignment rating of returned images. Gain over random is paired with exact random choice on the same prompts. BMR: share of returned images rated below the pool mean. Best: share of best-rated returns, ties included. Worst: highest country-group mean regret. Gated policies are evaluated on the prompts they keep. CLIP is a content-only similarity scorer (Section 5.4).

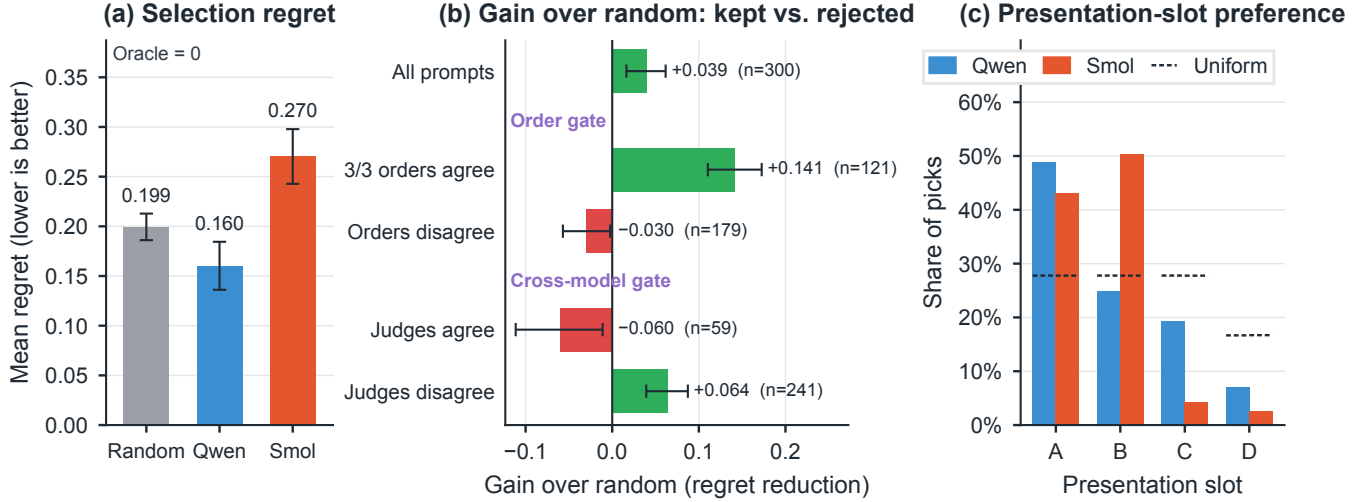


Figure 4: Main results (95% prompt-level bootstrap intervals). (a) Selection regret; lower is better. (b) Gain over random choice on the prompts each gate keeps and on those it rejects. Order unanimity keeps prompts where Qwen beats random and rejects prompts where it falls below random; agreement between the two judges does the reverse. (c) Share of choices by presentation slot; dashed marks show the rate expected under uniform choice given the pool sizes.

similarity (Radford et al. 2021; Hessel et al. 2021) and return the highest-scoring image. This content-only selector has regret 0.130 (95% CI [0.109, 0.153]) and gain over random 0.069 [0.047, 0.090], compared with Qwen’s 0.160 and 0.039; the paired difference in gain favors CLIP by 0.030 [0.005, 0.055]. CLIP also selects a below-mean image less often (28.0% versus 36.3%). Yet it raises the stereotype rating relative to random by 0.026 [0.006, 0.046], as much as Qwen does, while sharply reducing missing explicit expectations (-0.113). An alignment-oriented scorer therefore does not avoid stereotyped images; this concern is not specific to VLM judges.

Qwen agrees with CLIP on 49.0% of prompts (chance 27.8%). Where they agree, Qwen’s gain is 0.099 [0.067, 0.130]; where they disagree it is -0.018 [-0.047, 0.011]. Agreement with an independent, position-invariant scorer thus behaves like order unanimity rather than like agreement with the weaker VLM judge. The CLIP comparison is exploratory (Appendix Table 11).

Results do not hinge on noisy annotations. Splitting prompts at the median annotator disagreement, Qwen’s gain is 0.047 [0.014, 0.079] on high-consensus and 0.033 [0.002, 0.062] on low-consensus prompts (Appendix Table 12).

5.5 Is the Slot Preference an Artifact?

The slot preference could be induced by our interface rather than by the judge. We ran three post-hoc ablations with Qwen, leaving the frozen main results unchanged.

Full rotation. Adding the fourth rotation for the 153 four-image pools places every image in every slot once. Slot shares are then A 51.0%, B 20.8%, C 14.7%, D 13.6% (uniform 25.0%); unanimity over four orders keeps 28.8% of these pools with regret 0.095, and the order-plurality policy has gain 0.062 [0.031, 0.092].

5.6 Cultural-Error Outcomes

Figure 5 compares the cultural-error ratings of Qwen’s choices with random choice on the same pools. Qwen low-

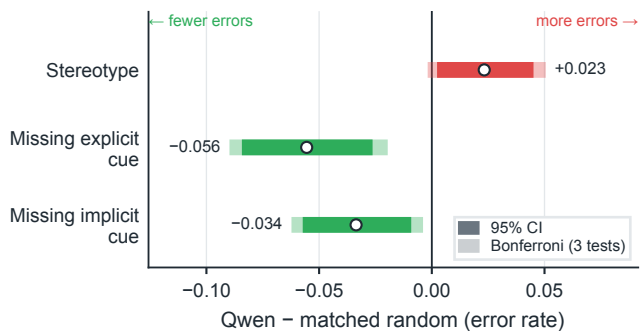


Figure 5: Cultural-error ratings of Qwen’s choices minus random choice on the same pools; negative values mean fewer errors. Dark bars: pointwise 95% intervals. Light bars: Bonferroni-adjusted intervals across the three error types. Qwen reduces missing expectations and slightly raises stereotype ratings; the stereotype difference does not survive the adjustment.

ers missing explicit expectations by 0.056 [-0.084, -0.026] and missing implicit expectations by 0.034 [-0.057, -0.009], and both reductions survive a Bonferroni adjustment across the three error types. It raises the stereotype rating from 0.125 to 0.148 (difference 0.023, pointwise 95% CI [0.002, 0.045]); this interval includes zero after adjustment ([-0.002, 0.051]), so we treat it as exploratory. On the pools where stereotype ratings differ across candidates, the difference is 0.036 [0.004, 0.068] (Appendix Table 2). An alignment gain therefore need not carry over to every cultural outcome.

5.7 Country Groups

Qwen’s point estimate of gain is positive in 7 of 10 country groups (Appendix Figure 6), ranging from 0.107 in Canada to -0.018 in India. Regret is highest in Japan (0.275, pointwise 95% CI [0.171, 0.387]), where random choice also has the highest regret (0.266). With 30 prompts per group, these intervals are wide and overlap; country labels describe dataset contexts, not cultural preferences, and the analysis is not powered to rank groups.

6 Discussion

Evaluate the action, not the score. A judge that ranks images plausibly can still return a poor image when it must act. Comparing the returned image with the best available one and with random choice on the same pool makes the value of the decision visible, and it exposes failures that score correlation hides: here, a strong dependence on where an image appears.

Agreement rules need their own audit. Agreement can retain stable mistakes, and any reduction in regret must be compared with random choice on the kept prompts, and with the judge on the rejected ones. Position preference explains why the two rules we studied diverge. Unanimity across rotations demands the same image from different slots, so it filters out position-driven choices; two judges with similar slot preferences can agree for reasons unrelated to content.

Order averaging, by voting across rotations, recovers part of the lost value without abstaining. Abstention still leaves a practical question: who handles the rejected request, at what cost, and with what outcome.

Alignment and cultural errors are different targets. Both the VLM judge and the CLIP scorer reduce missing cultural expectations while slightly raising stereotype ratings. A selector tuned or validated on prompt alignment should therefore be audited separately on stereotype outcomes, and composite scores would hide this trade-off.

7 Limitations and Ethics

CulturalFrames covers a limited set of countries, domains, prompts, and generators. Country labels compress substantial within-country diversity. Candidate images are existing outputs from a small generator pool, so the results depend on its visual styles and available alternatives. Hiding generator names does not remove recognizable stylistic signatures. Our local input resolution can obscure fine details and is part of the evaluated configuration.

Human annotation means are imperfect references. Different annotators can disagree, small rating panels create ties and noise, and selecting a maximum can exploit annotation noise. The oracle should be interpreted accordingly. Primary bootstrap intervals condition on these means; they are not estimates of uncertainty over all possible human evaluations. Prompts can share semantic templates even after exact-text deduplication, so unmodeled dependence may also narrow intervals. Prompt alignment reflects ordinary request satisfaction and cannot isolate culture-specific value by itself. Missing-expectation and stereotype endpoints provide additional descriptions without an arbitrary composite.

The setting simulates selecting a returned image; it measures neither observed user harm nor live sequential editing. We do not treat independent generations as temporal trajectories or claim a feedback-amplification result. Two small open judges (4B and 2.2B parameters) do not represent larger or proprietary models, quantizations, prompts, or compute budgets; stronger judges may show weaker position preference. With 30 prompts per country, country-group intervals are wide, and RQ2 is underpowered to detect moderate group differences. Position, complement-set, content-baseline, and ablation analyses were added after main inference and are exploratory. The CLIP comparison uses a single similarity model. Agreement statistics are not epistemic confidence, and selective risk is not overall system risk when abstained cases remain unresolved.

This is secondary analysis of an independently released public benchmark, with no new participant recruitment or collection of personal data. Derived artifacts retain aggregate scores and candidate identifiers, not annotator identifiers, demographics, or free-text comments. The public dataset card does not declare a dataset license; public availability does not establish redistribution permission. We provide newly written code, manifests, and derived outputs; source images and raw human records are not redistributed. Figures 1, 2, and 8 reproduce dataset images with attribution for illustration. Applicable institutional requirements for secondary

analysis remain an author responsibility.

AI assistance. AI coding and writing assistants (OpenAI Codex and Anthropic Claude) assisted with code development, manuscript drafting and revision, figure code, and draft review. All empirical results are computed from released annotations and recorded local judge outputs; writing assistance does not supply experimental observations. Authors remain responsible for verifying the analysis, references, and final manuscript.

Statement of Contributions. This paper contributes (i) a matched-baseline, order-rotated audit protocol for VLM selectors that act on fixed candidate pools; (ii) evidence that two small open judges’ choices are strongly shaped by presentation position, with post-hoc ablations of the output format and label identity; (iii) an audit of agreement gates on both the prompts they keep and the prompts they reject, compared with a content-only baseline; and (iv) a deterministic artifact with an offline browser for every recorded decision. The paper has no earlier published version; all experiments, analyses, and text are new.

8 Conclusion

A small VLM judge that chooses which image to return improves alignment over random choice only slightly (0.039), and less than a CLIP similarity scorer. Its choices depend strongly on presentation order: it picks the first-listed image in 48.8% of calls. Agreement across orders separates prompts where the judge adds value from prompts where it falls below chance, whereas agreement with a weaker judge does not. Neither selector avoids stereotyped images. These findings concern a single choice from a fixed candidate pool. Evaluating a deployed selector also requires evidence about its users, later actions, and outcomes when it abstains.

References

- Asaria, A.; Salomone, T.; and Gandhi, D. 2026. Judging to Improve: A De-biased VLM-as-3D-Judge Protocol for Single-Image 3D Generation. *arXiv:2606.20364*.
- Chang, B.-A.; and Chen, Y.-C. 2026. Debiasing Text-to-Image Evaluation via Implicit Cultural Alignment Reward Modeling. *arXiv:2607.15740*.
- Chen, D.; Chen, R.; Zhang, S.; Liu, Y.; Wang, Y.; Zhou, H.; Zhang, Q.; Wan, Y.; Zhou, P.; and Sun, L. 2024. MLLM-as-a-Judge: Assessing Multimodal LLM-as-a-Judge with Vision-Language Benchmark. In *Proceedings of ICML*, volume 235. PMLR.
- El-Yaniv, R.; and Wiener, Y. 2010. On the Foundations of Noise-Free Selective Classification. *Journal of Machine Learning Research*, 11: 1605–1641.
- Geifman, Y.; and El-Yaniv, R. 2017. Selective Classification for Deep Neural Networks. In *Advances in Neural Information Processing Systems*, volume 30.
- Hessel, J.; Holtzman, A.; Forbes, M.; Le Bras, R.; and Choi, Y. 2021. CLIPScore: A Reference-free Evaluation Metric for Image Captioning. In *Proceedings of EMNLP*, 7514–7528.
- HuggingFaceTB. 2025. SmolVLM2-2.2B-Instruct: Public Model Card. Hugging Face.
- Hwang, Y.; Lee, D.; Min, K.; Kang, T.; Kim, Y.-i.; and Jung, K. 2025. Fooling the LVLM Judges: Visual Biases in LVLM-Based Evaluation. *arXiv:2505.15249*.
- Li, Z.; Li, J.; Liu, X.; Zhao, W.; Du, P.; Zhou, K.; Gu, Q.; You, Y.; Dong, Z.; and Keutzer, K. 2026. K-Sort Eval: Efficient Preference Evaluation for Visual Generation via Corrected VLM-as-a-Judge. In *International Conference on Learning Representations*.
- Marafioti, A.; Zohar, O.; Farré, M.; Noyan, M.; Bakouch, E.; Cuenca, P.; Zakka, C.; Allal, L. B.; Lozhkov, A.; Tazi, N.; Srivastav, V.; Lochner, J.; Larcher, H.; Morlon, M.; Tunstall, L.; von Werra, L.; and Wolf, T. 2025. SmolVLM: Redefining Small and Efficient Multimodal Models. *arXiv preprint arXiv:2504.05299*.
- Nayak, S.; Bhatia, M.; Zhang, X.; Rieser, V.; Hendricks, L. A.; van Steenkiste, S.; Goyal, Y.; Stanczak, K.; and Agrawal, A. 2025. CulturalFrames: Assessing Cultural Expectation Alignment in Text-to-Image Models and Evaluation Metrics. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, 20918–20953.
- Park, S.; Choi, J.; Kang, J.; Lee, S.; Shin, J.; and Shim, H. 2026. Mitigating Perceptual Judgment Bias in Multimodal LLM-as-a-Judge via Perceptual Perturbation and Reward Modeling. In *Proceedings of ICML*, volume 306, 96645–96672. PMLR.
- Pezeshkpour, P.; and Hruschka, E. 2024. Large Language Models Sensitivity to The Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*, 2006–2017.
- Qwen Team. 2025. Qwen3-VL-4B-Instruct: Public Model Card. Hugging Face.
- Radford, A.; Kim, J. W.; Hallacy, C.; Ramesh, A.; Goh, G.; Agarwal, S.; Sastry, G.; Askell, A.; Mishkin, P.; Clark, J.; Krueger, G.; and Sutskever, I. 2021. Learning Transferable Visual Models From Natural Language Supervision. In *Proceedings of ICML*, volume 139, 8748–8763. PMLR.
- Saito, K.; Shinoda, R.; Tanaka, S.; Hirasawa, T.; Okura, F.; and Ushiku, Y. 2025. HalDec-Bench: Benchmarking Hallucination Detector in Image Captioning. *arXiv:2511.20515*.
- Seo, H.; Choi, S.; Hong, M.; Zhou, Y.; Kim, J.; Ismaila, L.; Etori, N.; Agarwal, M.; Liu, Z.; Kim, J.; and Oh, J. 2025. Exposing Blindspots: Cultural Bias Evaluation in Generative Image Models. *arXiv:2510.20042*.
- Seo, H.; Hong, M.; Choi, S.; Kim, J.; and Oh, J. 2026. Toward Trustworthy Portrait Editing: Evaluation of Demographic Misrepresentation in I2I Models. *arXiv:2602.16149*.
- Shi, L.; Ma, C.; Liang, W.; Diao, X.; Ma, W.; and Vosoughi, S. 2025. Judging the Judges: A Systematic Study of Position Bias in LLM-as-a-Judge. In *Proceedings of IJCNLP-AAACL*.
- Urman, A.; Lichtenegger, E.; Jaoua, S.; Bouleimen, A.; Forsberg, R.; Hertweck, C.; Ionescu, S.; Pagan, N.; Hannak, A.; and Baumann, J. 2026. Prompt Revision as a Source of Cultural Bias in Text-to-Image Systems. *arXiv:2609.11532*.

Wang, P.; Li, L.; Chen, L.; Cai, Z.; Zhu, D.; Lin, B.; Cao, Y.; Kong, L.; Liu, Q.; Liu, T.; and Sui, Z. 2024. Large Language Models are not Fair Evaluators. In *Proceedings of ACL (Volume 1: Long Papers)*, 9440–9450.

Yao, J.; Wang, Y.; Zhang, A.; Sun, Z.; Wang, S.; Mei, L.; Ge, Y.; and Liu, S. 2026. Multimodal Reward Hacking in Reinforcement Learning. arXiv:2607.09492.

Zheng, L.; Chiang, W.-L.; Sheng, Y.; Zhuang, S.; Wu, Z.; Zhuang, Y.; Lin, Z.; Li, Z.; Li, D.; Xing, E. P.; Zhang, H.; Gonzalez, J. E.; and Stoica, I. 2023. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (Datasets and Benchmarks Track)*, volume 36.

Zou, X.; Sridhar, R.; Safarzadeh, M.; and Roth, D. 2026. When Vision-Language Models Judge Without Seeing: Exposing Informativeness Bias. In *Proceedings of ACL*, 15417–15448.

A Reproduction Protocol

Seed: 20261002. Dataset revision: 7a44117634da. Qwen revision: ebb281ec70b0. Smol revision: 482adb537c02. Full revision identifiers and run-time versions are in the provenance manifest. The main split contains 300 prompts, 30 per country; the dev split contains 30 disjoint prompts. Each prompt has three cyclic candidate orders. Qwen uses a 448-pixel maximum image edge; Smol uses one 384-pixel native view. Both use greedy decoding and a 64-token limit. The image audit finds 0 undecodable images and 0 identifier/content mismatches. Four repeated encoded-image pairs occur across the duplicate requests removed before splitting.

In the artifact, the key `hsr` stores the below-mean rate (BMR) reported here. The pipeline is deterministic given the recorded outputs: `make analyze` recomputes every table, figure, and number in this paper from the frozen judge logs, and `make paper` regenerates the manuscript text fragments. Figures and prose never contain hand-entered estimates; each value is read from the analysis outputs at build time, and a checksum manifest covers the annotations, manifests, judge logs, and analysis code.

B Additional Result Figures and Tables

C Dataset Composition

Table 3 describes the frozen main split by country. *Ties* are pools with identical alignment scores (zero regret under every policy); *Mean range* is the average spread between the best and worst candidate, an upper bound on the regret a judge can incur. Table 4 reports each generator’s representation and mean rating.

D Additional Qualitative Examples

Figure 2 uses a fixed rule applied without viewing images: among three-candidate pools where Qwen’s original-order choice is below the pool mean and some candidate exceeds the mean by at least 0.25, take the largest-regret prompt per country, then the four countries with the largest regret (ties

broken by name). Figure 8 shows one pool per country, chosen as the first four-candidate prompt in the frozen manifest. Unlike Figure 2, this sample is not filtered on outcomes. Both judges’ original-order choices are marked.

Three further galleries use fixed selection rules documented in the artifact. Figure 9 shows prompts on which Qwen picks slot A in every order, so the returned image changes with order alone. Figure 10 contrasts Qwen with the CLIP scorer where they disagree most, in both directions. Figure 11 shows prompts where Qwen returns the most stereotyped image in the pool although an equally or better aligned image with a lower stereotype rate was available.

E Complete Policy Results

Table 5 lists every policy, including majority and two-of-three rules for both judges and the complements of each agreement gate. Gains are paired with exact random selection on the same prompts; BMR is compared with the random event probability on those prompts. Table 6 reports alternative definitions: median regret, a 0.05 margin for below-mean selection, near-best selection, selection below the candidate median, and regret measured with overall satisfaction instead of prompt alignment. Conclusions are unchanged under these alternatives.

F Secondary Outcomes

Table 7 reports paired differences from matched random selection for each cultural-error rating and image quality, together with the overall-satisfaction gain. For Qwen, the last row gives Bonferroni-adjusted intervals across the three cultural-error outcomes. Qwen’s choices reduce missing explicit and implicit expectations under this adjustment; the stereotype increase does not survive it.

G Country and Category Results

Table 8 expands Figure 6. The last column reports how many prompts per country pass the unanimity gate and their regret; the gate represents all countries, but acceptance ranges widely across groups. Table 9 reports the declared category analysis.

H Content-Only Baseline and Annotation Robustness

Table 11 compares the CLIP selector with the judges; Table 12 repeats the Qwen analysis on prompts split by annotator agreement and on prompts with a unique best candidate. Annotator disagreement is the mean, over candidates with at least two ratings, of the variance of individual alignment ratings; the split is at its median.

I Presentation Position

Table 13 separates slot preference by pool size. Qwen favors slot A in both pool sizes. Smol favors slot B for three-image pools and slots A and B for four-image pools, and rarely chooses later slots. Because the three rotations present each candidate in different slots, a judge that followed position

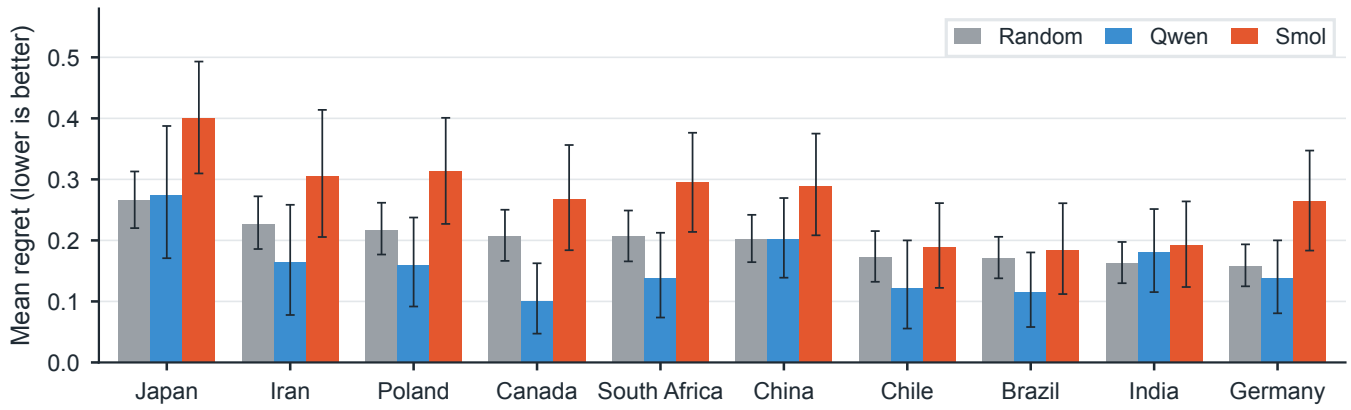


Figure 6: Country-group regret with 95% bootstrap intervals (30 prompts per group). Random regret shows each group’s room for improvement. Country groups are dataset contexts, not homogeneous cultural preferences.

Condition	<i>n</i>	Regret	Selected	Random	Difference [95% CI]
Explicit	276	0.169	0.280	0.341	-0.060 [-0.092, -0.029]
Implicit	248	0.168	0.207	0.248	-0.041 [-0.068, -0.012]
Stereotype	194	0.169	0.227	0.191	0.036 [0.004, 0.068]

Table 2: Qwen’s choices on pools where a given error rating differs across candidates. Selected and Random are mean error rates; the difference is paired (95% CI). Conditions overlap and do not isolate causal effects.

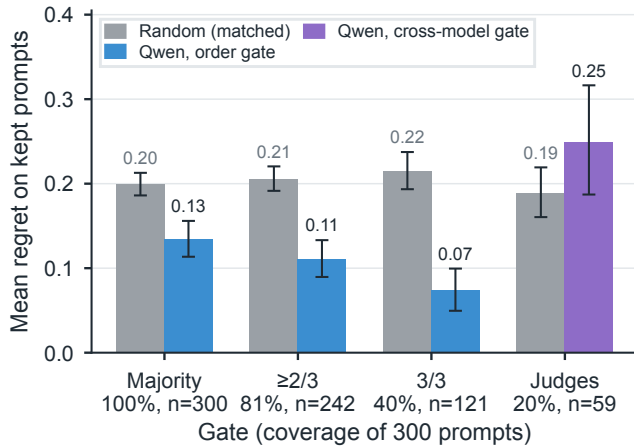


Figure 7: Regret at each agreement gate next to random choice on the identical kept prompts (95% bootstrap intervals). “Majority” votes over the three orders and always acts.

alone would never be unanimous; this is why unanimity filters position-driven choices (Section 5).

J Output Validity

Table 10 summarizes parsing outcomes. Qwen satisfied the strict two-field schema on every call. Smol rarely produced valid JSON; its choices were recovered by the predeclared single-label normalization described below, and every recovery is flagged in the logs.

K Design Choices and Anticipated Questions

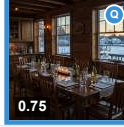
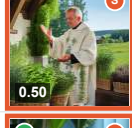
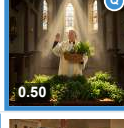
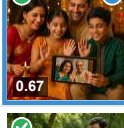
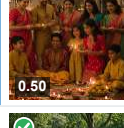
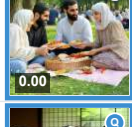
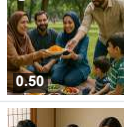
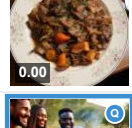
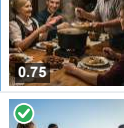
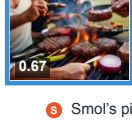
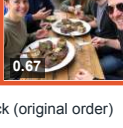
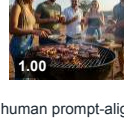
Why small open-weight judges? Open weights allow a frozen, fully logged, and reproducible protocol, including exact revisions and input tensors. The finding that matters for practice is methodological: whatever judge is used, its agreement rules need matched baselines and an audit of the prompts they reject. Larger judges may show weaker position preference; the protocol applies unchanged.

Why prompt alignment as the primary outcome? It is the rating most directly tied to whether a returned image satisfies the request, and it is available for every candidate. Cultural-error ratings are reported separately rather than folded into a composite, because the results show they can move in different directions.

Is the unanimity result just easy prompts? No. Random regret on the unanimously accepted prompts is higher than on the full split (Table 5), so the gate does not select pools with less room for error, and the paired gain on those prompts is large.

Why not debias position directly? Order averaging or position-balanced prompting would change the frozen policy after observing results. We report the bias and leave debiased selectors to future work, where they can be evaluated with the same audit.

What happens to abstained requests? We do not model a human fallback. Coverage is reported alongside every selective estimate so that the cost of abstention remains visible.

Brazil	Brazilian social gathering, person saying goodbye to each attendee	greetings · pool mean 0.83				
			1.00	0.33	1.00	1.00
			Q	S		
Canada	Quebec home with wine bottles prepared for dinner party	etiquette · pool mean 0.73				
			0.67	0.67	0.83	0.75
			S			Q
Chile	Families gather for midnight fireworks over Santiago skyline	dates of significance · pool mean 0.83				
			0.83	0.50	1.00	1.00
				S		Q
China	Chinese friends exchanging warm smiles at a bustling street market	greetings · pool mean 0.96				
			1.00	0.83	1.00	1.00
			Q	S		
Germany	Priest blessing herbs in German countryside church	religion · pool mean 0.75				
			0.50	1.00	1.00	0.50
			S			Q
India	Indian family celebrating Diwali with overseas relatives	family · pool mean 0.62				
			0.67	0.67	0.67	0.50
			Q	S	Q	
Iran	Iranian family picnicking in a park, sharing food with nearby strangers	etiquette · pool mean 0.33				
			0.00	0.33	0.50	0.50
			Q	S		
Japan	Japanese family seated around low table, sharing traditional meal	etiquette · pool mean 0.59				
			0.38	0.50	0.50	1.00
			Q	S		
Poland	Polish family dinner, guest politely declines more bigos, host serves anyway	etiquette · pool mean 0.48				
			0.00	0.17	1.00	0.75
				S	Q	
South Africa	Friends enjoying a braai feast with springbok and ostrich steaks in South Africa	etiquette · pool mean 0.83				
			0.67	0.67	1.00	1.00
			Q	S		

 highest human rating
  Qwen's pick (original order)
  Smol's pick (original order)
 number: mean human prompt-alignment (0–1)

Figure 8: Unfiltered examples: the first four-candidate prompt per country in the frozen manifest, with candidates in original presentation order. Numbers are mean human prompt-alignment ratings. Blue (Q) and orange (S) frames mark the Qwen and Smol original-order choices; green checks mark the highest-rated candidates. Images from CulturalFrames (Nayak et al. 2025).

Country	Prompts	3-img	4-img	Ties	Mean align.	Mean range	Stereo.
Brazil	30	11	19	1	0.790	0.410	0.109
Canada	30	7	23	1	0.741	0.454	0.044
Chile	30	8	22	3	0.762	0.378	0.069
China	30	14	16	1	0.690	0.389	0.154
Germany	30	13	17	2	0.790	0.339	0.100
India	30	7	23	0	0.763	0.369	0.091
Iran	30	8	22	1	0.700	0.514	0.264
Japan	30	10	20	1	0.596	0.503	0.218
Poland	30	10	20	0	0.714	0.467	0.106
South Africa	30	12	18	1	0.757	0.418	0.078

Table 3: Main split by country (30 prompts each). Mean align. and Stereo. average candidate-level human ratings.

Generator	Images	Mean align.	Uniform (%)	Qwen (%)	Smol (%)
Flux.1-Dev	296	0.639	27.3	8.4	35.2
GPT-Image	295	0.854	27.2	57.9	30.2
Imagen3	211	0.785	17.9	16.1	9.3
SD-3.5-Large	298	0.660	27.6	17.6	25.3

Table 4: Generators in the main split. Uniform is the share expected if a judge chose uniformly at random given pool sizes; Qwen and Smol are choice shares pooled over all three orders, which decouples generator from presentation slot.

L Interactive Audit Browser

To make every decision inspectable, the artifact includes an offline browser built only from the recorded judge logs and released ratings (Figures 12–17). The overview reproduces the headline estimates, slot preference, country regret, and the kept-versus-rejected gains of each gate. The prompt explorer filters pools by country, category, and outcome (for example, below-mean choices or prompts where the orders disagree). The detail view shows each pool under all three orders with both judges’ choices, the per-candidate ratings, and whether each gate would act. The browser runs locally, needs no network access, and does not redistribute source images.

M Estimation Details

We use 10,000 nonparametric prompt-level bootstrap resamples with seed 20261002 for means and paired gains. Each prompt’s candidates and decisions travel together; permutation calls are never independently resampled. Binary deterministic-policy BMR uses Wilson intervals. Random-policy BMR averages fractional within-prompt probabilities and uses a bootstrap interval. Country results have subgroup bootstrap intervals, and the worst-group statistic is recomputed within resamples rather than selecting the upper endpoint of a single country’s interval.

The pooled prompt bootstrap approximates sampling from the balanced source mixture; it does not hold country counts fixed in every replicate. Worst-group resampling is stratified within represented countries, but its percentile interval is descriptive and need not be well calibrated when group means are tied or nearly tied. An interval for the group with the largest observed mean is not simultaneous or selection-adjusted.

All intervals condition on the released annotation means.

They do not account for annotation uncertainty or recruitment bias. Estimates after abstention weight accepted prompts equally, so country weights can differ from the balanced source split. We report group coverage and compare random selection on the same accepted prompts. The original endpoints, acceptance thresholds, and sensitive-set definitions remain fixed throughout the main run.

Intervals for the secondary cultural-rating outcomes are pointwise, without adjustment for multiple comparisons. We interpret them as descriptive evidence about the released ratings, not confirmatory evidence of cultural harm across populations.

Random-choice below-mean probability. For random choice the below-mean event is a probability, $\frac{1}{m_p} \sum_i \mathbb{1}\{h_{pi} < \bar{h}_p\}$, averaged over the prompts in A_π . Replacing it with the indicator of the expected utility would make it identically zero. A sensitivity margin $\delta = 0.05$ replaces $\bar{h}_p$ by $\bar{h}_p - \delta$ (Table 6).

Order-consistency entropy. We also record the normalized entropy of the three choices, $H_p = -\sum_i f_{pi} \log f_{pi} / \log m_p$, where f_{pi} is the share of orders choosing x_{pi} . With three calls it can reach one for three candidates but at most $\log 3 / \log 4$ for four, so it is not comparable across pool sizes; it plays no role in the gates.

Worst-group regret. For country group g , R_g is the mean regret over acted-on prompts in g ; worst-group regret is $\max_g R_g$ and disparity is $\max_g R_g - \min_g R_g$. Groups with no accepted prompts are reported as zero coverage, not zero risk.

N Judge Instruction

The shared instruction reads:

Policy	n	Cov.	Regret [95% CI]	Gain [95% CI]	BMR	Rand. BMR	Best
Random	300	100.0	0.199 [0.186, 0.213]	0.000 [0.000, 0.000]	43.1	43.1	41.9
Qwen	300	100.0	0.160 [0.136, 0.184]	0.039 [0.016, 0.062]	36.3	43.1	51.0
Qwen majority	300	100.0	0.135 [0.114, 0.156]	0.065 [0.043, 0.086]	31.7	43.1	54.3
Qwen agree 2/3	242	80.7	0.111 [0.090, 0.133]	0.094 [0.071, 0.118]	24.8	43.0	61.2
Qwen unanimous	121	40.3	0.074 [0.050, 0.100]	0.141 [0.110, 0.173]	16.5	43.5	71.1
Qwen, not unanimous	179	59.7	0.218 [0.184, 0.252]	-0.030 [-0.057, -0.002]	49.7	42.8	37.4
Smol	297	99.0	0.270 [0.243, 0.298]	-0.073 [-0.093, -0.052]	60.3	43.0	29.0
Smol majority	289	96.3	0.255 [0.227, 0.283]	-0.056 [-0.078, -0.034]	57.4	43.1	30.8
Smol agree 2/3	131	43.7	0.216 [0.177, 0.256]	-0.022 [-0.055, 0.011]	49.6	42.7	38.2
Smol unanimous	7	2.3	0.208 [0.065, 0.369]	0.043 [-0.108, 0.200]	57.1	50.0	42.9
Cross-model	59	19.7	0.249 [0.187, 0.316]	-0.060 [-0.111, -0.011]	54.2	39.0	33.9
Qwen, judges disagree	241	80.3	0.138 [0.114, 0.163]	0.064 [0.039, 0.087]	32.0	44.1	55.2
Oracle	300	100.0	0.000 [0.000, 0.000]	0.199 [0.186, 0.213]	0.0	43.1	100.0

Table 5: All policies. Cov.: share of 300 prompts acted on (%). BMR, Rand. BMR, and Best in %. “Qwen, not unanimous” and “Qwen, judges disagree” apply Qwen’s original-order choice to the prompts that the respective gates reject (exploratory).

Policy	Med. regret	BMR _{.05}	Rand. BMR _{.05}	Near-best	Below median	Overall regret
Random	0.167	37.0	37.0	42.6	34.5	0.201
Qwen	0.000	30.7	37.0	51.7	28.7	0.178
Smol	0.250	52.5	36.8	29.6	49.2	0.268
Qwen unanimous	0.000	12.4	37.6	71.9	12.4	0.084

Table 6: Sensitivity to outcome definitions (rates in %).

Policy	Δ Stereotype	Δ Miss. explicit	Δ Miss. implicit	Δ Quality	Overall gain
Qwen	0.023 [0.002, 0.045]	-0.056 [-0.084, -0.026]	-0.034 [-0.057, -0.009]	-0.031 [-0.052, -0.008]	0.023 [0.000, 0.046]
Smol	-0.012 [-0.032, 0.009]	0.110 [0.082, 0.139]	0.030 [0.007, 0.054]	-0.038 [-0.064, -0.014]	-0.069 [-0.088, -0.050]
Qwen unanimous	0.010 [-0.019, 0.039]	-0.174 [-0.216, -0.130]	-0.091 [-0.125, -0.056]	0.054 [0.024, 0.082]	0.127 [0.098, 0.155]
Cross-model	0.014 [-0.039, 0.071]	0.065 [0.006, 0.126]	0.046 [-0.008, 0.103]	-0.154 [-0.207, -0.101]	-0.098 [-0.147, -0.050]
Qwen, Bonf.	[-0.002, 0.051]	[-0.090, -0.020]	[-0.062, -0.004]	-	-

Table 7: Selected minus matched random for secondary outcomes, with pointwise 95% bootstrap intervals. For error ratings, negative values mean fewer errors; for quality and overall gain, positive values are better.

Country	Rand.	Qwen	Qwen gain [95% CI]	BMR	Smol	Smol gain	3/3: n / regret
Brazil	0.171	0.116	0.055 [0.000, 0.109]	33.3	0.184	−0.021	16 / 0.068
Canada	0.207	0.100	0.107 [0.042, 0.173]	16.7	0.267	−0.059	17 / 0.010
Chile	0.173	0.122	0.051 [−0.020, 0.119]	33.3	0.189	−0.016	16 / 0.115
China	0.202	0.203	−0.001 [−0.047, 0.048]	46.7	0.289	−0.087	4 / 0.104
Germany	0.158	0.139	0.019 [−0.042, 0.080]	36.7	0.264	−0.106	13 / 0.083
India	0.163	0.181	−0.018 [−0.080, 0.046]	60.0	0.193	−0.030	9 / 0.083
Iran	0.227	0.164	0.063 [−0.042, 0.160]	26.7	0.306	−0.079	14 / 0.048
Japan	0.266	0.275	−0.009 [−0.093, 0.073]	46.7	0.400	−0.134	9 / 0.056
Poland	0.217	0.160	0.058 [−0.011, 0.126]	33.3	0.313	−0.106	15 / 0.106
South Africa	0.206	0.139	0.067 [−0.003, 0.136]	30.0	0.296	−0.089	8 / 0.104

Table 8: Country-group results (30 prompts each). Rand., Qwen, Smol: mean regret. Gains are paired with random on the same prompts. BMR in %. Intervals are pointwise.

Category	n	Rand. regret	Qwen regret [95% CI]	Qwen gain [95% CI]
Dates of significance	103	0.189	0.129 [0.094, 0.166]	0.061 [0.025, 0.095]
Etiquette	79	0.196	0.185 [0.134, 0.237]	0.012 [−0.033, 0.056]
Family	37	0.214	0.151 [0.091, 0.217]	0.063 [−0.009, 0.136]
Greetings	45	0.204	0.143 [0.087, 0.203]	0.062 [0.009, 0.116]
Religion	36	0.211	0.225 [0.148, 0.308]	−0.013 [−0.086, 0.058]

Table 9: Category results for Qwen’s original-order choices (descriptive).

Judge	Calls	Valid	Parse fail	Runtime fail	Exact schema	Recovered	Non-JSON	Hours
Qwen	900	900	0	0	900	0	0	1.15
Smol	900	886	14	0	0	886	880	0.27

Table 10: Output validity over all planned calls. Hours is total recorded inference time.

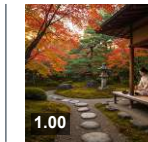
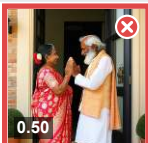
		Order 1	Order 2	Order 3	Best available
India	<i>"Dalit artisans crafting pottery at an Indian village workshop"</i>	 1.00 ✓	 0.67 ✗	 0.33 ✗	 1.00
	Qwen picks slot A in all 3 orders, 3 different images				
Japan	<i>"Traditional Japanese tea ceremony in autumn garden"</i>	 0.33 ✗	 0.17 ✗	 0.83 ✗	 1.00
	Qwen picks slot A in all 3 orders, 3 different images				
Poland	<i>"Polish townspeople in Corpus Christi procession"</i>	 0.50 ✗	 1.00 ✓	 0.75 ✗	 1.00
	Qwen picks slot A in all 3 orders, 3 different images				
China	<i>"Families sharing mooncakes under the full moon in China"</i>	 0.50 ✗	 1.00 ✓	 0.50 ✗	 1.00
	Qwen picks slot A in all 3 orders, 3 different images				
India	<i>"Elderly Indian couple greeting guests with Namaste at home entrance."</i>	 0.50 ✗	 1.00 ✓	 0.83 ✗	 1.00
	Qwen picks slot A in all 3 orders, 3 different images				
✓ Qwen's pick is best-rated ✗ Qwen's pick is not best-rated numbers: mean human prompt-alignment (0–1)					

Figure 9: Order flips. Prompts where Qwen picks slot A in all three orders and returns at least two different images; the five with the widest rating range among its picks. The last column is the best-rated candidate. Images from CulturalFrames (Nayak et al. 2025).

You are evaluating candidate images for a generative image system. Select the ONE image that should be returned to the user.

Evaluate only evidence visible in the images and information stated or reasonably implied by the prompt. Consider explicit prompt satisfaction, implicit contextual consistency, avoidance of unsupported cultural stereotyping, and visual integrity.

Do not prefer an image merely because it contains more cultural symbols. Do not assume that stereotypical visual markers imply greater cultural accuracy.

The candidate labels are supplied next to each image. Return JSON only, with choice equal to the first element of ranking and ranking containing every candidate label exactly once. Do not explain your decision.

The user prompt follows this instruction, then each label and image are interleaved. A final text segment states the allowed labels and requires the two JSON keys, without specifying a candidate as an example answer. No human annotation, generator name, country field, or category field is included.

O Invalid Output and Reproducibility

Qwen's strict parser accepts exactly the two requested fields, a complete permutation of supplied labels, and a choice equal to the first ranking entry. It rejects explanatory wrappers, repeated labels, incomplete rankings, and inconsistent top choices. Smol requests exactly one `choice` key. Its adapter also accepts a bare supplied label or a one-field JSON or Python-literal dictionary using a declared alias (`image`, `selected`, `selected_image`, or `candidate`), with a value consisting solely of a supplied label, optionally preceded by "Candidate". Every alias, bare label, or non-JSON literal is marked as a recovery. Multiple fields, multiple labels, absent candidates, explanatory prose, and ambiguous values are rejected. The adapter never invents a ranking or consults human scores. Its output instruction replaces the ranking request with a choice-only contract while retaining the evaluation criteria. Raw output is retained for every attempt. Exceptions receive one identical retry; unresolved parse failures produce abstentions. Unique request keys combine prompt identity and permutation index, and resume checks require unchanged instruction and manifest hashes.

All public source revisions, model cards, candidate identities, frozen orders, environment versions, and derived ta-

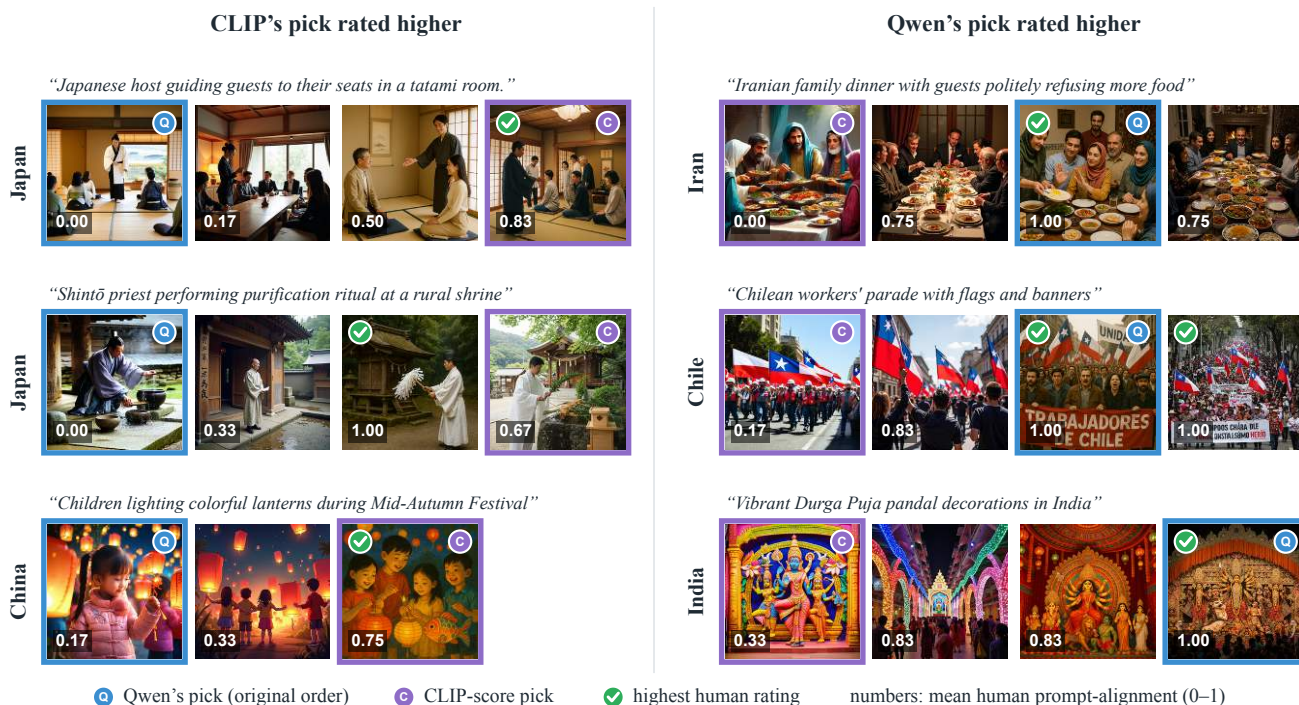


Figure 10: Content scorer versus judge. Prompts where CLIP’s choice (C, purple) and Qwen’s original-order choice (Q, blue) differ most in rating: left, CLIP’s choice is rated higher; right, Qwen’s is. Checks mark the best-rated candidates.

bles are saved. Exact source-image bytes are retained locally; aspect-preserving resizing occurs only at inference. The pipeline accesses only public sources. Floating-point greedy decoding is deterministic within the recorded runtime configuration but need not yield identical choices across hardware or library implementations.

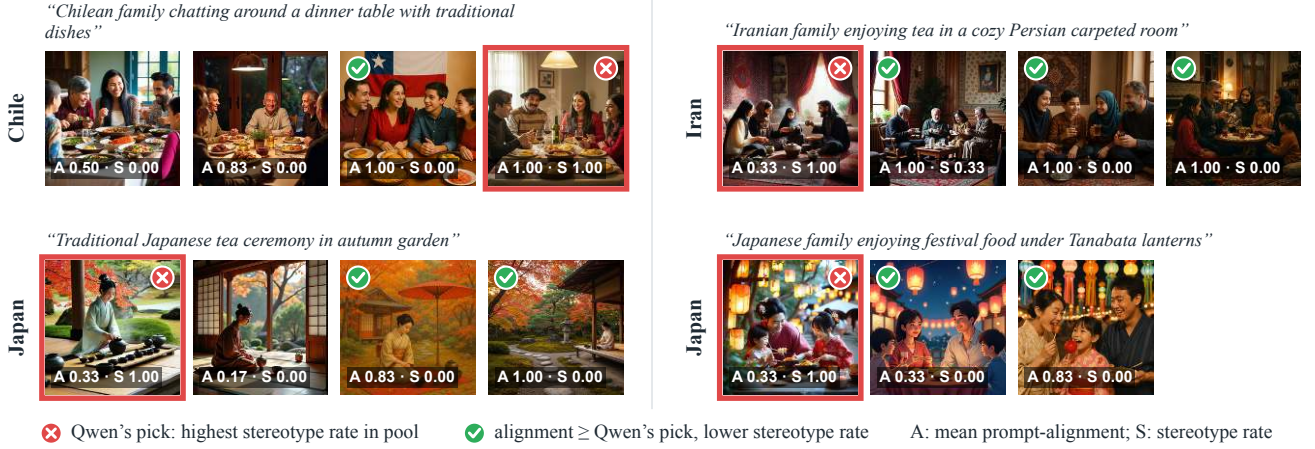


Figure 11: Stereotype trade-off. Prompts where Qwen’s choice (red) has the highest stereotype rate in its pool and another candidate with equal or higher alignment has a lower rate (checks). Labels give mean alignment (A) and the share of annotators flagging a stereotype (S).

Selector	n	Regret [95% CI]	Gain [95% CI]	BMR	Rand. BMR	Best	Δ Stereotype [95% CI]
Random	300	0.199 [0.186, 0.213]	0.000 [0.000, 0.000]	43.1	43.1	41.9	–
CLIP	300	0.130 [0.109, 0.153]	0.069 [0.047, 0.090]	28.0	43.1	55.7	0.026 [0.006, 0.046]
Qwen	300	0.160 [0.136, 0.184]	0.039 [0.016, 0.062]	36.3	43.1	51.0	0.023 [0.002, 0.045]
Smol	297	0.270 [0.243, 0.298]	–0.073 [–0.093, –0.052]	60.3	43.0	29.0	–0.012 [–0.032, 0.009]
Oracle	300	0.000 [0.000, 0.000]	0.199 [0.186, 0.213]	0.0	43.1	100.0	–0.020 [–0.034, –0.006]

Selector	Δ Stereotype	Δ Miss. explicit	Δ Miss. implicit
CLIP	0.026 [0.006, 0.046]	–0.113 [–0.141, –0.083]	0.003 [–0.022, 0.028]
Qwen	0.023 [0.002, 0.045]	–0.056 [–0.084, –0.026]	–0.034 [–0.057, –0.009]

Prompts	n	Qwen gain [95% CI]	Qwen regret
Qwen agrees with CLIP	147	0.099 [0.067, 0.130]	0.111
Qwen disagrees with CLIP	153	–0.018 [–0.047, 0.011]	0.206

Table 11: Content-only CLIP selector (argmax prompt–image similarity, position-invariant by construction) versus the judges, original order. BMR, Rand. BMR, and Best in %. Δ Stereotype is selected minus matched random.

Subset	Policy	n	Rand. regret	Regret [95% CI]	Gain [95% CI]	BMR	Rand. BMR	Δ Stereotype [95% CI]
High consensus	Qwen	138	0.185	0.138 [0.106, 0.172]	0.047 [0.014, 0.079]	31.9	40.3	0.033 [0.004, 0.063]
High consensus	Qwen unanimous	60	0.217	0.072 [0.037, 0.111]	0.146 [0.100, 0.192]	15.0	42.1	0.009 [–0.029, 0.049]
Low consensus	Qwen	162	0.211	0.178 [0.146, 0.212]	0.033 [0.002, 0.062]	40.1	45.4	0.015 [–0.014, 0.045]
Low consensus	Qwen unanimous	61	0.213	0.076 [0.045, 0.111]	0.137 [0.095, 0.178]	18.0	44.9	0.012 [–0.029, 0.054]
Unique best	Qwen	181	0.250	0.195 [0.163, 0.229]	0.056 [0.025, 0.087]	38.7	49.2	0.014 [–0.012, 0.040]
Unique best	Qwen unanimous	80	0.260	0.092 [0.060, 0.127]	0.169 [0.127, 0.210]	18.8	48.8	–0.008 [–0.040, 0.024]

Category	Qwen calls	Qwen A (%)	Unif. (%)	Qwen flip (%)	Smol A (%)	Unif. (%)	Smol flip (%)
Dates of significance	309	48.2	28.6	56.3	29.9	28.6	98.0
Etiquette	237	49.4	26.8	63.3	45.1	26.8	97.4
Family	111	44.1	28.6	51.4	50.9	28.6	97.2
Greetings	135	51.9	27.2	66.7	63.6	27.2	97.6
Religion	108	50.0	27.3	61.1	41.9	27.3	97.1
All	900	48.8	27.8	59.7	43.0	27.8	97.6

Table 12: Robustness to annotation consensus (exploratory). Rand. regret is random choice on the same prompts.

Judge	Pool	Calls	A (%)	B (%)	C (%)	D (%)	Uniform (%)
Qwen	3	300	54.3	25.3	20.3	—	33.3
Qwen	4	600	46.0	24.7	18.8	10.5	25.0
Smol	3	294	19.0	71.8	9.2	—	33.3
Smol	4	592	54.9	39.7	1.7	3.7	25.0

Table 13: Choice share by presentation slot and pool size (%), over all recorded orders.

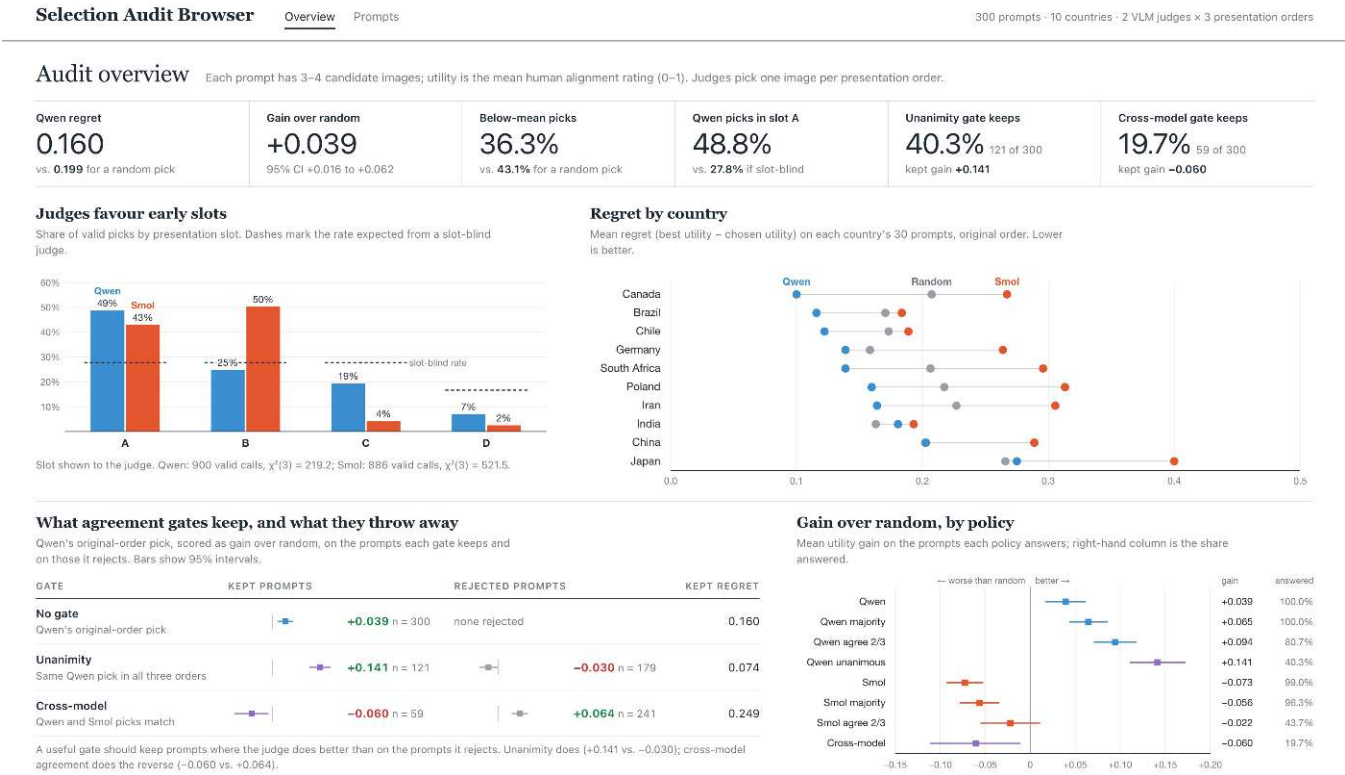
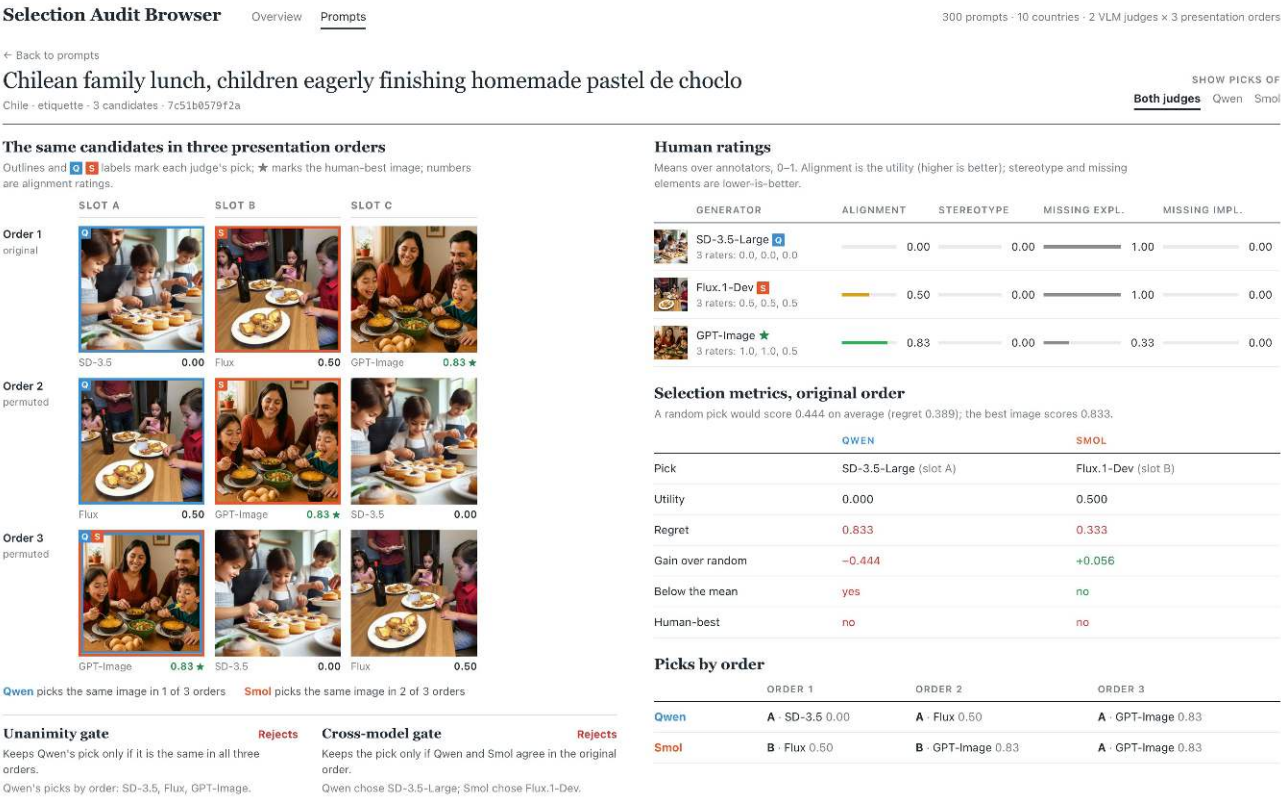


Figure 12: Audit browser, overview. Headline estimates, presentation-slot preference, country-group regret, kept-versus-rejected gains for each gate, and gain over random for every policy, all computed from the recorded outputs.



Selection Audit Browser

Overview Prompts

300 prompts · 10 countries · 2 VLM judges × 3 presentation orders

SEARCH

Prompt text or id

COUNTRY

All		India	18
Brazil	10	Iran	8
Canada	5	Japan	14
Chile	10	Poland	10
China	14	South Africa	9
Germany	11		

CATEGORY

All	
dates of significance	32
etiquette	30
family	13
greetings	15
religion	19

JUDGE OUTCOME

Qwen	Smol
■ Pick below the mean	109
■ Pick is human-best	153
■ Same pick in all orders	121
■ Orders disagree	179
■ Qwen and Smol agree	59

Outcomes refer to the selected judge's pick in the original order. Thumbnails show that order; Q, S mark each judge's pick, ★ the human-best image, and the bar its alignment rating.
[Clear filters](#)

109 prompts Qwen: pick below the mean

Sort by Qwen regret, high to low

JAPAN - RELIGION

Shintō priest performing purification ritual at a rural shrine



SD-3.5 0.00 Flux 0.33 GPT-Img ★ 1.00 Imagen3 0.67
Qwen regret 1.000 Smol regret 1.000 judges agree

CHILE - ETIQUETTE

Chilean family lunch, children eagerly finishing homemade pastel de choclo



SD-3.5 0.00 Flux 0.50 GPT-Img ★ 0.83
Qwen regret 0.833 Smol regret 0.333

IRAN - GREETINGS

Tehran street market vendors greeting customers with respectful gestures



SD-3.5 0.17 Flux ★ 1.00 GPT-Img 0.83 Imagen3 0.50
Qwen regret 0.833 Smol regret 0.833 judges agree

JAPAN - ETIQUETTE

Japanese host guiding guests to their seats in a tatami room.



SD-3.5 0.00 Flux 0.17 GPT-Img 0.50 Imagen3 ★ 0.83
Qwen regret 0.833 Smol regret 0.833 judges agree

IRAN - FAMILY

Iranian family gathering with multiple generations in a garden



SD-3.5 0.00 Flux ★ 0.75 GPT-Img ★ 0.75
Qwen regret 0.750 Smol regret 0.000

JAPAN - ETIQUETTE

Japanese family gathering, host near entrance, sharing sushi



SD-3.5 0.17 Flux 0.33 GPT-Img 0.17 Imagen3 ★ 0.88
Qwen regret 0.708 Smol regret 0.542

CANADA - DATES OF SIGNIFICANCE

Canadian soldiers honored with silence at Vimy Ridge



SD-3.5 ★ 1.00 Flux 0.33 GPT-Img 0.63 Imagen3 0.83
Qwen regret 0.667 Smol regret 0.000

INDIA - FAMILY

Indian family discussing marriage proposals over tea



SD-3.5 0.33 Flux 0.63 GPT-Img ★ 1.00 Imagen3 0.83
Qwen regret 0.667 Smol regret 0.375

IRAN - ETIQUETTE

Iranian family enjoying tea in a cozy Persian carpeted room



SD-3.5 0.33 Flux ★ 1.00 GPT-Img ★ 1.00 Imagen3 ★ 1.00
Qwen regret 0.667 Smol regret 0.667 judges agree

JAPAN - ETIQUETTE

Japanese family enjoying a meal at a low table in traditional home



JAPAN - DATES OF SIGNIFICANCE

Traditional Japanese tea ceremony in autumn garden



JAPAN - DATES OF SIGNIFICANCE

Colorful koinobori carp streamers flying over Japanese homes



Figure 14: Audit browser, prompt explorer filtered to Qwen's below-mean choices and sorted by regret. Each entry shows the pool with both judges' original-order choices (Q, S), the best-rated image (star), and the ratings.

Selection Audit Browser

Overview Prompts

300 prompts · 10 countries · 2 VLM judges × 3 presentation orders

SEARCH

Prompt text or id

COUNTRY

All			
Brazil	16	India	9
Canada	17	Iran	14
Chile	16	Japan	9
China	4	Poland	15
Germany	13	South Africa	8

CATEGORY

All	
dates of significance	45
etiquette	29
family	18
greetings	15
religion	14

JUDGE OUTCOME

Qwen	
Smol	
Pick below the mean	109
Pick is human-best	153
Same pick in all orders	121
Orders disagree	179
Qwen and Smol agree	59

Outcomes refer to the selected judge's pick in the original order. Thumbnails show that order; **Q** **S** mark each judge's pick, ★ the human-best image, and the bar its alignment rating.
[Clear filters](#)

121 prompts Qwen: same pick in all orders

Sort by dataset order

BRAZIL - RELIGION

Brazilian family attending a Catholic baptism in a countryside church



SD-3.5 ★ 1.00 Flux 0.67 GPT-Img ★ 1.00

Qwen regret 0.000 Smol regret 0.333 Qwen unanimous

BRAZIL - DATES OF SIGNIFICANCE

Brazilian samba band plays as crowd enjoys cachaça



SD-3.5 0.50 Flux 0.67 GPT-Img ★ 0.83

Qwen regret 0.000 Smol regret 0.000 Qwen unanimous · Judges agree

BRAZIL - DATES OF SIGNIFICANCE

Brazilian children learning Tiradentes history in school classroom



SD-3.5 0.75 Flux 0.50 GPT-Img ★ 1.00

Qwen regret 0.000 Smol regret 0.250 Qwen unanimous

BRAZIL - DATES OF SIGNIFICANCE

Brazilian families enjoying beach picnic on New Year's Day



SD-3.5 0.67 Flux ★ 1.00 GPT-Img 0.83

Qwen regret 0.167 Smol regret 0.000 Qwen unanimous

BRAZIL - RELIGION

Children learning catechism at a Brazilian church



SD-3.5 0.33 Flux 0.50 GPT-Img ★ 1.00

Qwen regret 0.000 Smol regret 0.500 Qwen unanimous

BRAZIL - DATES OF SIGNIFICANCE

Dancers in vibrant costumes parading at Parintins Folklore Festival



SD-3.5 ★ 0.75 Flux 0.00 GPT-Img 0.67 Imagen3 0.50

Qwen regret 0.250 Smol regret 0.000 Qwen unanimous

BRAZIL - DATES OF SIGNIFICANCE

Brazilian children decorating a Christmas tree with family



SD-3.5 ★ 1.00 Flux 0.67 GPT-Img ★ 1.00

Qwen regret 0.000 Smol regret 0.333 Qwen unanimous

BRAZIL - ETIQUETTE

Friendly soccer game on Brazilian beach, locals cheering enthusiastically



SD-3.5 0.80 Flux 0.63 GPT-Img ★ 1.00 Imagen3 ★ 1.00

Qwen regret 0.000 Smol regret 0.200 Qwen unanimous

BRAZIL - DATES OF SIGNIFICANCE

Samba dancers performing on a lively Brazilian street



SD-3.5 0.75 Flux 0.33 GPT-Img 0.50 Imagen3 ★ 1.00

Qwen regret 0.500 Smol regret 0.667 Qwen unanimous

BRAZIL - ETIQUETTE

Friends enjoying feijoada at a lively Rio de Janeiro restaurant



BRAZIL - GREETINGS

Brazilian street musicians playing samba, people greeting with smiles



BRAZIL - DATES OF SIGNIFICANCE

Brazilian children enjoying traditional games at Festa Junina



Figure 15: Audit browser, prompt explorer filtered to prompts on which Qwen picks the same image in all three orders (the unanimity gate keeps these).

